

Motor Datasheet: WxC D230L90

Operating Point: Continuous | Temperature: 20 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	200.0	mm
Machine Inner Diameter	120.0	mm
Machine Thickness / Width	50.6	mm
Air Gap	1.00	mm
Fill Factor (input)	0.65	-
Slot Fill Factor (calculated)	0.41	-
Slot Width (tooth to tooth)	9.358	mm
Available Coil Space	4.679	mm
Assumed Coil Width (simulation)	3.904	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	5.03	kg
Rotor Mass	701.3	g
Stator Mass	4.33	kg
Rotor Structure Included	No	-
Rotor Moment of Inertia (active elements only)	0.0047	kg·m²

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	161.155	A
Supply Current (Peak)	227.907	A
Id (Peak)	-0.000	A
Iq (Peak)	227.907	A
Phase Advance Angle	0.0	°elec
Phase Resistance	0.844	mOhm
D-Axis Inductance (Ld)	3.552	μH
Q-Axis Inductance (Lq)	3.875	μH
Terminal Inductance (Ltt = Ld + Lq)	7.427	μH
PM Flux Linkage	6.465	mWb
Frequency at operating point (electrical)	500.000	Hz
Winding Configuration	wye (star)	-
DC Supply Voltage	48.000	V
Power Factor	0.982	-

Performance

Parameter	Value	Unit
Torque (Continuous)	23.25	Nm
Speed	3000.0	rpm
BackEMF Constant (Ke, phase peak)	6.8	mV/rpm
Motor Constant (Km, only resistive losses)	2.87	Nm/√W
Specific Motor Constant (Km / mass)	570.5	mNm/√W/kg
Torque Constant (Kt, peak, no-load)	96.7	mNm/A
Torque Ripple	5.4	%

Saturation	0.0	%
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Losses

Parameter	Value	Unit
Total Losses	284.0	W
Resistive Losses	65.7	W
Iron Core Losses	166.9	W
Magnet Eddy Current Losses	51.4	W
Rotor Structure Eddy Current Losses	0.0	W
Winding Eddy Current Losses	0.0	W

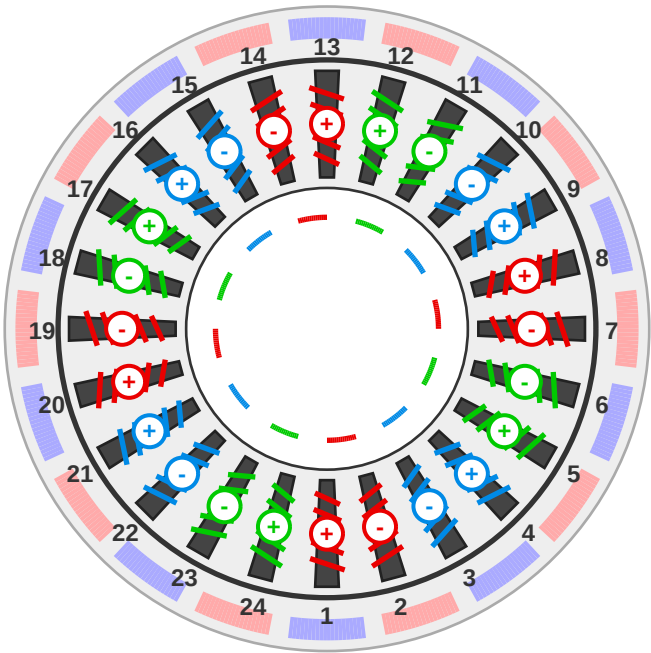
Configuration

Parameter	Value	Unit
Number of Stators	2	-
Number of Rotors	1	-
Number of Slots	24	-
Pole Pairs	10	-
Number of Parallel Paths	2	-
Number of Turns per Tooth (designed incl. parallel paths)	2.0	-
Number of Phase Turns (all phases in series)	16.0	-

Winding Configuration

Schema: AabBCcaABbcCAabBCcaABbcC

Concentrated Winding | Balanced | Cogging: 120/rev | Ripple: 120/rev



Materials

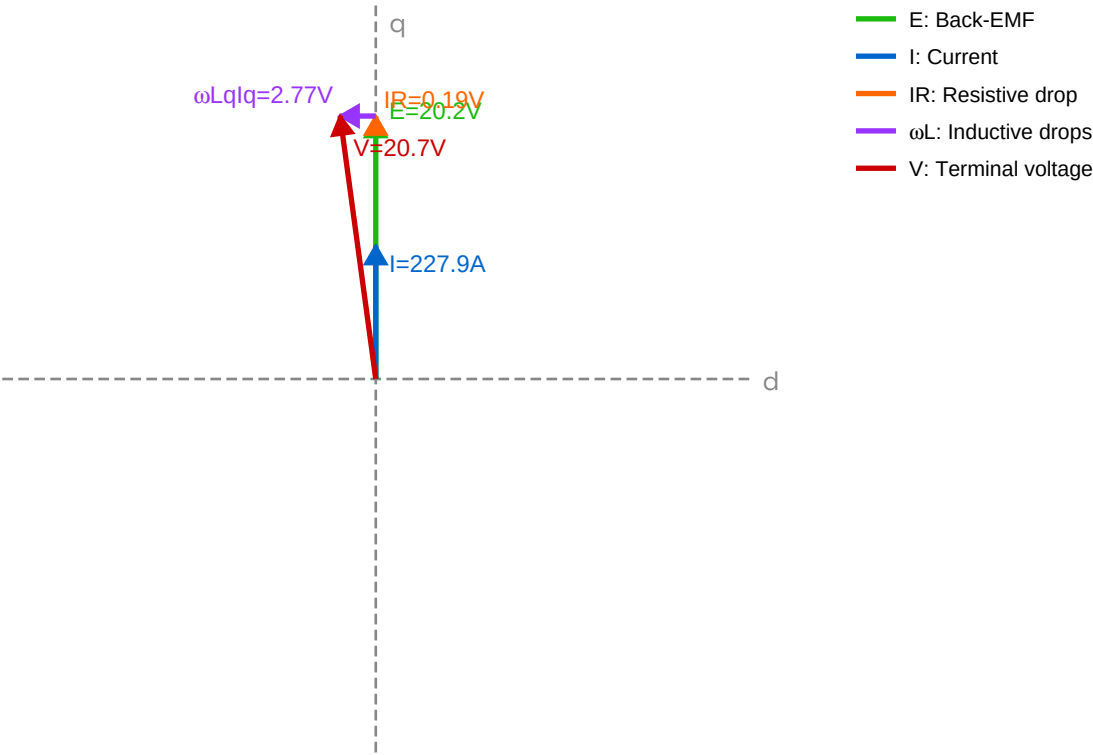
Component	Material	
Stator Core	B35A250-BaoSteel	-
Magnets	BMN-48SH	-

Voltage Breakdown at Operating Point (with integer turns)

Parameter	Value	Unit
Back-EMF Voltage (Phase, Peak)	20.245	V
Resistive Drop (Phase, Peak)	0.192	V
Inductive Drop Ld (Peak)	0.000	V
Inductive Drop Lq (Peak)	2.775	V
Inductive Drop Total (Peak)	2.775	V
Terminal Voltage (Phase, Peak)	20.735	V

Back-EMF Voltage (Line-to-Line, Peak)	35.065	V
Resistive Drop (Line-to-Line, Peak)	0.333	V
Inductive Drop (Line-to-Line, Peak)	4.806	V
Terminal Voltage (Line-to-Line, Peak)	35.914	V

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms: $\text{Peak} = \text{RMS} \times \sqrt{2}$.
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection: $V_{LL} = V_{\text{phase}} \times \sqrt{3}$.